

Algebraic Properties of Real Numbers:

Algebraic Properties are the basic rules that describe how mathematical operations such as addition, subtraction, multiplication & division behave.

1- Close Law: For $a, b \in \mathbb{R}$.

$$a + b \in \mathbb{R}$$

eg not hold in $\mathbb{N}$, $a \cdot b \in \mathbb{R}$.

2 Commutative:

$$a + b = b + a$$

$$a \cdot b = b \cdot a$$

3) Associative

$$a + (b + c) = (a + b) + c$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

4) Identity

For $a \in \mathbb{R}$, $\exists 0 \in \mathbb{R}$ s.t

$$a + 0 = 0 + a = a$$

& $\exists 1 \in \mathbb{R}$ s.t

5) Inverse

$$a \cdot 1 = 1 \cdot a = a$$

For $a \in \mathbb{R}$, $\exists$ ^{unique} $-a \in \mathbb{R}$ s.t $a + (-a) = (-a) + a = 0$

" " $\exists \frac{1}{a} \in \mathbb{R}$ s.t $a \cdot (\frac{1}{a}) = (\frac{1}{a}) \cdot a = 1$.

6) Distributive

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

Inequalities:

A statement of the form $x < y$ or $(y > x)$.

"The mathematical expression containing inequality symbols ($<$ or $>$)".
* Solution of inequality is interval

Properties:

Let $a, b, c \in \mathbb{R}$.

i) one & only one is true
 $a < b$ or $b < a$ or $a = b$.] Trichotomy.

ii) If $a < b$ & $b < c$ then $a < c$] Transitivity

iii) If $a < b$ then $a + c < b + c$ } Addition
 $a - c < b - c$ } Subtraction

iv) If $a < b$
 $\begin{cases} ac < bc \\ ac > bc \end{cases}$ $\begin{matrix} c > 0 \text{ (preserve)} \\ c < 0 \text{ (reverse)} \end{matrix}$ } Multiplication / Division
 $\downarrow$
 $a < b ; c > 0$
 $\frac{a}{c} < \frac{b}{c}$

v) If $a < b$ (Reciprocal)
reverse $\frac{1}{a} > \frac{1}{b}$ both a & b have same sign
 $\frac{a}{c} < \frac{b}{c} ; c < 0$
 $\frac{a}{c} > \frac{b}{c}$

preserve $\frac{1}{a} < \frac{1}{b}$ (opposite signs)

Q: & solution of inequality is interval,

What is the solution of equation?

Transitive

Transitive

Substitution

Multiplication

Reciprocal

→ Interval:

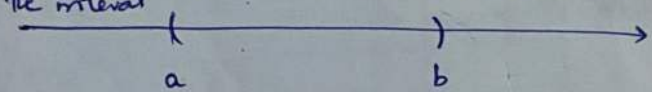
A subset of real line is called an interval if it contains at least two numbers and contains all the real numbers lying between these two.

⇒ Bounded - Intervals (a and b are fixed)

Let a and b be fixed real numbers with $a < b$.

- (i) The open interval (a, b) with end points a and b is the set of all real numbers x such that $a < x < b$.

a does not belong to the interval b does not belong to the interval



A horizontal number line with arrows at both ends. Two points are marked on the line: 'a' on the left and 'b' on the right. Above the line, between 'a' and 'b', there is an open parenthesis '(' at 'a' and a closing parenthesis ')' at 'b'. The line continues to the right with an arrow.

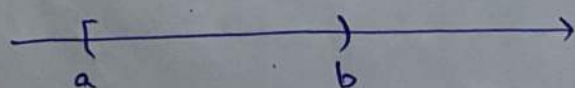
- (ii) The closed interval $[a, b]$ with end points a and b is the set of all real numbers x such that $a \leq x \leq b$.

'b' belongs to interval. 'a' belongs to the interval



A horizontal number line with arrows at both ends. Two points are marked on the line: 'a' on the left and 'b' on the right. Above the line, between 'a' and 'b', there is a bracket '[' at 'a' and a bracket ']' at 'b'. The line continues to the right with an arrow.

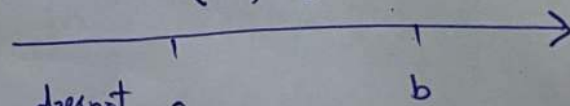
- (iii) The half closed / half - open interval with end point a and b
- $[a, b)$ $(a, b]$



A horizontal number line with arrows at both ends. Two points are marked on the line: 'a' on the left and 'b' on the right. Above the line, at 'a' there is a bracket '[' and at 'b' there is a parenthesis ')'. The line continues to the right with an arrow.

a belongs to interval

b does not belong to the interval



A horizontal number line with arrows at both ends. Two points are marked on the line: 'a' on the left and 'b' on the right. Above the line, at 'a' there is a parenthesis '(' and at 'b' there is a bracket ']'. The line continues to the right with an arrow.

does not a belongs to the interval

b belongs to the interval

* Note:

① $\rightarrow$ A closed interval contains its end points, but an open interval does not.

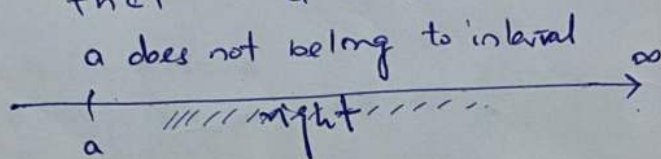
② $\rightarrow$ A half-open / half-closed contains one of its end points, but not the other.

→ Unbounded- Intervals:

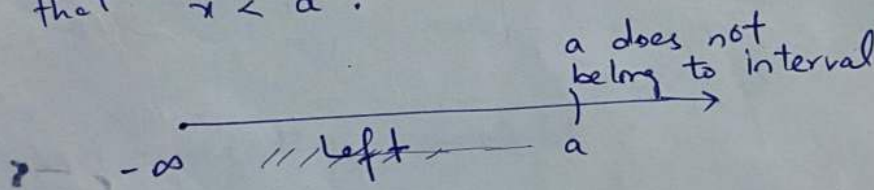
An interval which extend indefinitely to the right or left, are written with the help of special symbols $+\infty$ and $-\infty$.

one end-point is fixed. let a be fixed real no.

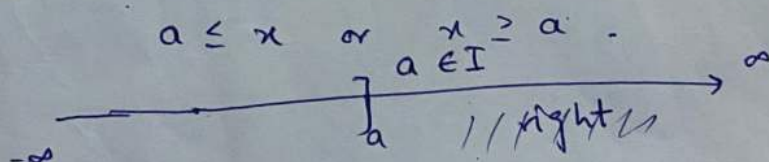
(i) $(a, +\infty)$ is set of all real number x such that $a < x$. $x > a$



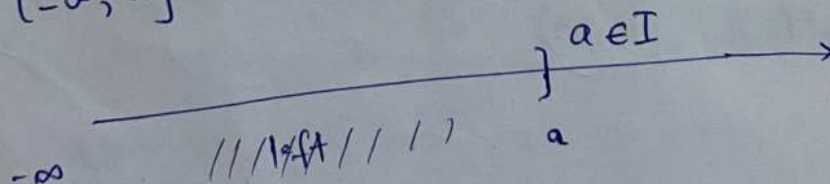
(ii) $(-\infty, a)$ is set of all real numbers x such that $x < a$.



(iii) $[a, \infty)$



(iv) $(-\infty, a]$



$$x \leq a$$

Q: what about set of real no's?

How this set is written in interval form.

$$x^2 - 3 < 1$$

$$x^2 - 4 < 0 \quad (\text{Make R.H.S equal to zero}).$$

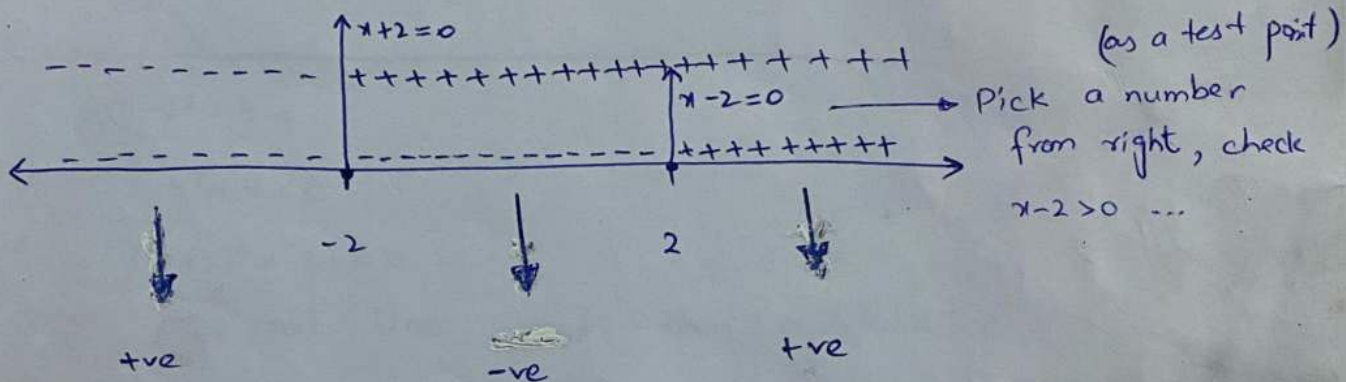
$$(x+2)(x-2) < 0.$$

* We find the negative product of $(x+2)$ and $(x-2)$.

Firstly, locate the zero points, that is

$$x+2=0 \quad \text{when} \quad x=-2$$

$$x-2=0 \quad \text{when} \quad x=2$$



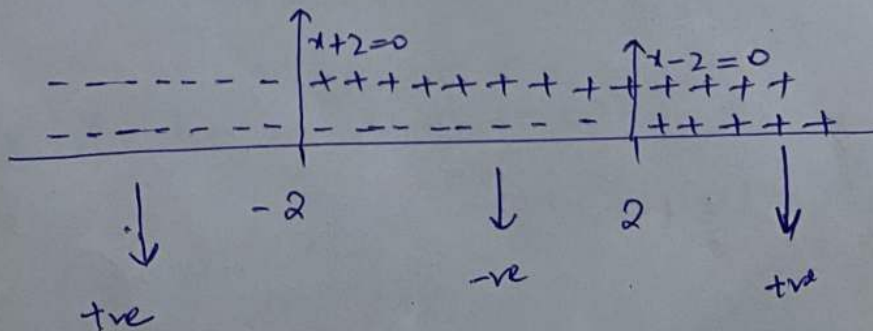
we want negative product

So, the solution is $(-2, 2)$

$$x^2 - 3 > 1$$

$$x^2 - 4 > 0 \quad (\text{Make R.H.S zero})$$

$$(x+2)(x-2) > 0$$



we want the product of $(x+2)$ and $(x-2)$ (11)

So, the solution is

$$(-\infty, -2) \cup (2, \infty)$$

If consider equality sign in above.

$$(-\infty, -2] \cup [2, \infty)$$

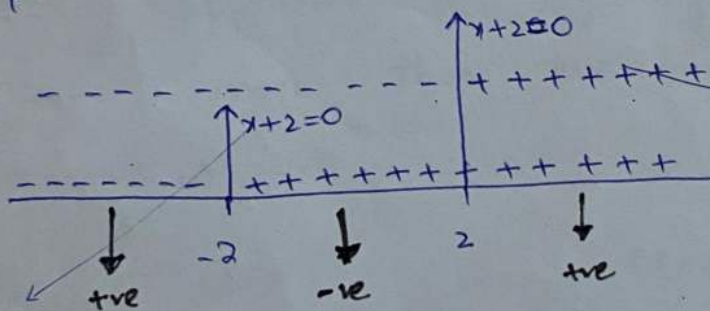
$\Rightarrow$ Equality sign will change open intervals to half open / half closed.

$$x^2 - 3 < 1$$

$$x^2 - 4 < 0$$

$$(x+2)(x-2) < 0$$

Locate zero pt in real line and then mark the signs.



*pick a point from right, check
 $0+2=10 > 0$

$$-1+2=1 > 0$$

$$\text{So, S.S} = (-2, 2)$$

$$\text{For } x^2 - 3 > 0$$

$$\text{S.S} = (-\infty, -2) \cup (2, \infty)$$

$$\text{Like } x^2 - 3 \leq 1 \quad \text{S.S} = [-2, 2]$$

$$x^2 - 3 \geq 1 \quad \text{S.S} = (-\infty, -2] \cup [2, \infty)$$

Alternate Approach:

$$(x+2)(x-2) < 0$$

Either $x+2 < 0$

or $x-2 < 0$

If $x+2 < 0$ and $(x-2) > 0$

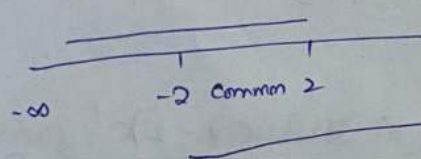
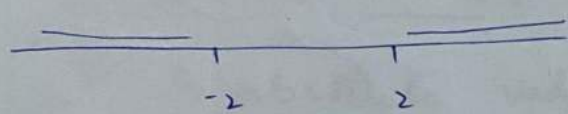
or $x-2 < 0$ and $x+2 > 0$

$x < -2$ and $x > 2$

or $x < 2$ and $x > -2$

$(-\infty, -2) \cap (2, \infty)$

or $(-\infty, 2) \cap (-2, \infty)$



$\Rightarrow \phi$

$\cup (-2, 2)$

$\Rightarrow S.S = \phi \cup (-2, 2) = (-2, 2)$

This (second) approach is confusing & lengthy b/c if we have more than two intervals or like

$(x+2)(x-2) > 0$

$(x+2) < 0$ & $(x-2) < 0$

or $(x-2) > 0$ & $(x+2) > 0$

$x > -2$

$x < 2$

Second approach has chance of mistake b/c of confusion but choice is yours.

Q- $\frac{3}{x-2} > 1$; $x \neq 2$

$\rightarrow$ How to solve by cross-multiplying, by taking reciprocals we do these b/c $x-2$ involve 'x' variable due to which $x-2$ has dual sign

like when $x > 2$ $x-2 > 0$ (ve)
 & $x < 2$ $x-2 < 0$ (-ve)

So this disturb the

So this produce confusion.

$$\frac{3}{x-2} \geq 1$$

(Make R.H.S = 0)

$$\Rightarrow \frac{3}{x-2} - 1 \geq 0$$

$$\frac{3 - (x-2)}{x-2} \geq 0$$

$$\frac{3 - x + 2}{x-2} \geq 0$$

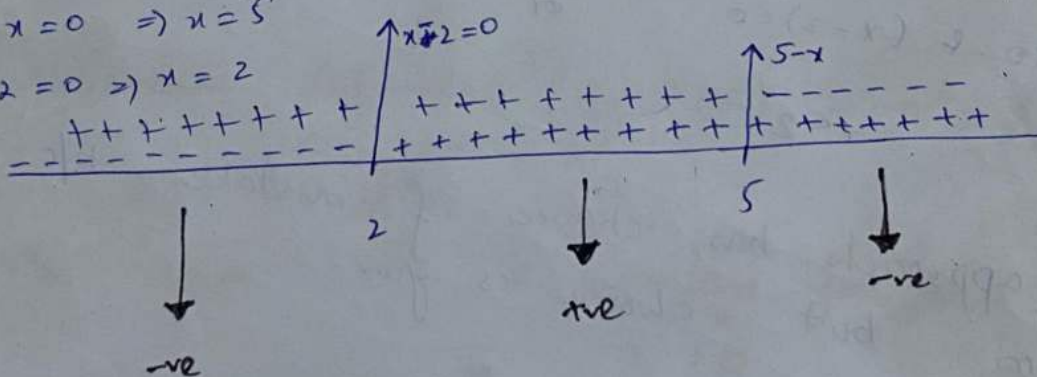
$$\frac{5-x}{x-2} \geq 0$$

$$(5-x) \left(\frac{1}{x-2} \right) \geq 0$$

+ sign of $x+2$ & $\frac{1}{x+2}$ has same nature.

$$5-x=0 \Rightarrow x=5$$

$$x-2=0 \Rightarrow x=2$$



we need +ve product

$$\text{So, S.S} = (2, 5]$$

2 is included b/c $\frac{1}{x-2}$

Q: $\frac{3x-1}{2x+5} < 4$, $(-\infty, 4)$

Q: $2x-1 < x+3$, $(-3/2, \infty)$

Q: $-\frac{x}{3} < 2x+1$, $(1, 11/5]$

Q: $\frac{6}{x-1} \geq 5$, 1

Absolute Value:

The absolute value of a nos 'x', denoted by $|x|$, is defined by the formula:

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Geometrically, absolute value of x is the distance from x to '0' which is always positive. i.e.

$$|x| \geq 0 \quad \& \quad |x| = 0 \Leftrightarrow x = 0$$

Imp^{*}

An alternate definition of $|x|$ is

$$|x| = \sqrt{x^2}$$

Never write $\sqrt{x^2} = x$ unless you don't know $x > 0$

⑤ Absolute Value Properties:

(14)

(i) $|a| \geq 0$

(ii) $|-a| = |a|$ A no. or its additive inverse have same absolute value.

(iii) $|ab| = |a||b|$ The absolute value of the product is the product of absolute values.

(iv) $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ Same is true for quotient.

$\mathcal{I}_{mp}^*$
(v) $|a+b| \leq |a| + |b|$ $\rightarrow$ triangular inequality

Examples: $|-3+5| = |2| = 2 < |-3| + |5|$
 $= 3 + 5$
 $= 8$

$|3+5| = |8| = |3| + |5|$

$|-3-5| = |-8| = 8 = |-3| + |-5|$
 $= 3 + 5$
 $= 8$

$|x| < a$ means distance b/w x and "0" is less than a i.e. x must be b/w a and $-a$

Absolute value and intervals:

If a is any positive no., then

(1) $|x| = a \Leftrightarrow x = \pm a$

(2) $|x| < a \Leftrightarrow -a < x < a$

(3) $|x| > a \Leftrightarrow x > a \text{ or } x < -a$

(4) $|x| \leq a \Leftrightarrow -a \leq x \leq a$ (5) $|x| \geq a \Leftrightarrow x \geq a \text{ or } x \leq -a$

$$-5x > 6$$

or

$$-5x < -12$$

Multiplying $(-\frac{1}{5})$

$$-5(-\frac{1}{5})x < 6(-\frac{1}{5})$$

$$\Rightarrow x < -\frac{6}{5}$$

or

$$x > \frac{12}{5}$$

$$(-\infty, -\frac{6}{5})$$

$$\cup (\frac{12}{5}, \infty)$$

$$Q: \left| \frac{2x-1}{3x+2} \right| < 5$$

* Most generalized Question
include all properties of ineq
& 11.

using ②

$$\frac{2x-1}{3x+2} < 5 \text{ and}$$

$$-5 < \frac{2x-1}{3x+2} \text{ and } \frac{2x-1}{3x+2} < 5$$

($\frac{2x-1}{3x+2} > -5$
Make ~~both~~ side zero).

$$\frac{2x-1}{3x+2} + 5 > 0 \text{ and } \frac{2x-1}{3x+2} - 5 > 0$$

$$\frac{2x-1+5(3x+2)}{3x+2} > 0 \text{ and } \frac{2x-1-5(3x+2)}{3x+2} > 0$$

$$\frac{2x-1+15x+10}{3x+2} > 0$$

$$\text{and } \frac{2x-1-15x-10}{3x+2} > 0$$

$$\frac{17x+9}{3x+2} > 0$$

and

$$\frac{-13x-11}{3x+2} > 0$$

and

$$\frac{-13x-11}{3x+2} \left(\frac{1}{3x+2} \right) > 0$$

Absolute valued - Ineq

$$|3-5x|-2 > 7$$

$$|3-5x| > 9$$

using ③

$$3-5x > 9 \text{ or } 3-5x < -9$$

$$-5x$$

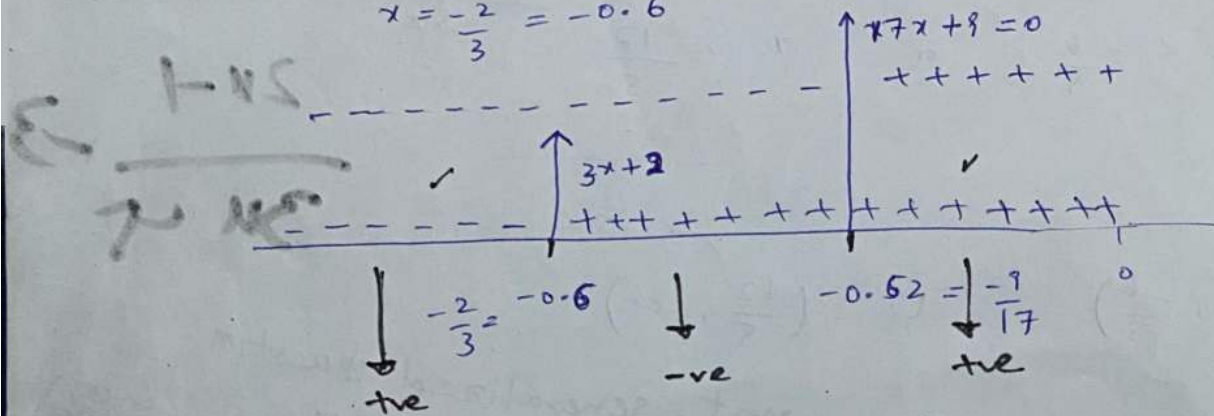
$$\frac{2x-1}{3x+2} > 3$$

* Each inequality has separate real line.

$$17x + 9 \left(\frac{1}{3x+2} \right) > 0$$

zero pts: $x = -\frac{9}{17} = -0.52$,

$$x = -\frac{2}{3} = -0.6$$



* we need the product

$$S.S \left(-\infty, -\frac{2}{3} \right) \cup \left(-\frac{9}{17}, \infty \right)$$

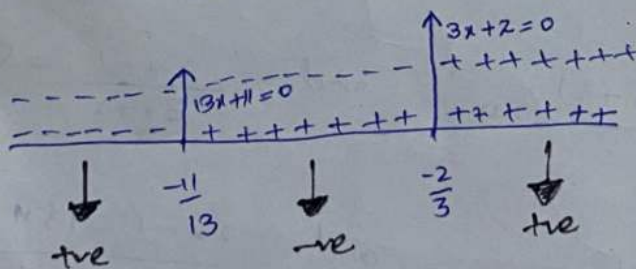
Now for $\frac{13x+11}{3x+2} > 0$

$$-(13x+11) \left(\frac{1}{3x+2} \right) > 0$$

zero pt: $-(13x+11) = 0$;

$$3x+2=0$$

$$13x+11 \Rightarrow x = -\frac{11}{13} = -0.8 ; \quad x = -\frac{2}{3} = -0.6$$

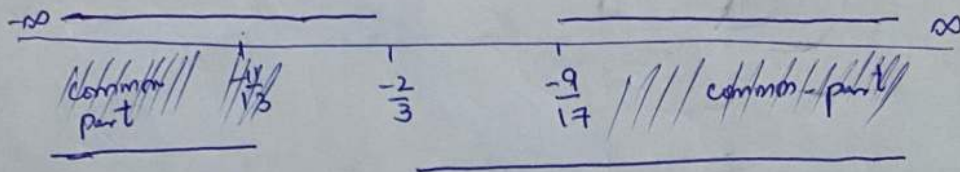


$$\Rightarrow S.S \left(-\infty, -\frac{11}{13} \right) \cup \left(-\frac{2}{3}, \infty \right)$$

Therefore
 $S.S = \left\{ \left(-\infty, -\frac{2}{3} \right) \cup \left(-\frac{9}{17}, \infty \right) \right\}$ and $\left\{ \left(-\infty, -\frac{11}{13} \right) \cup \left(-\frac{2}{3}, \infty \right) \right\}$

Now for intersection we & consider the real line. (10) (15)

if there is union not need consider the real line. 16/17



$$S.S = (-\infty, -\frac{11}{13}) \cup (-\frac{9}{17}, \infty).$$

→ $|x| > \sqrt{x}$ (Is we discuss power or inequality??)

To deal such inequalities, we sketch (need) the graph.
To study the graph, we firstly discuss the function.

* (Portion related to real numbers, inequalities, interval and absolute value is on Appendix A.1 of Thomas calculus Ed-14).

Function:

rule (command)

$$f: X \rightarrow Y$$

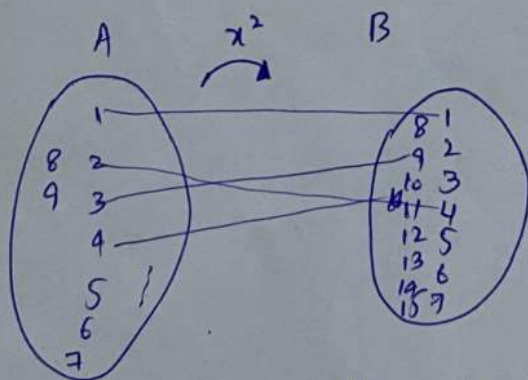
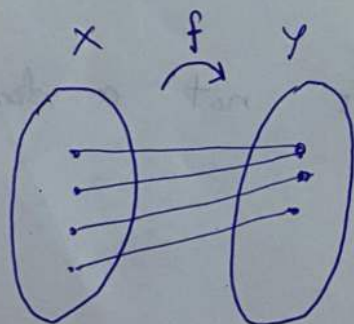
A function from a set X to a set Y is a rule that assigns a unique value $f(x)$ in Y to each x in X .

Domain:

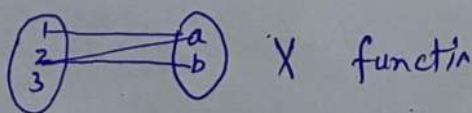
The set X of all allowed (permissible) / possible input values is called domain of function, written as $\text{Dom } f$.

Range:

The set of all output values $y = f(x) \in Y$ is called range of function.



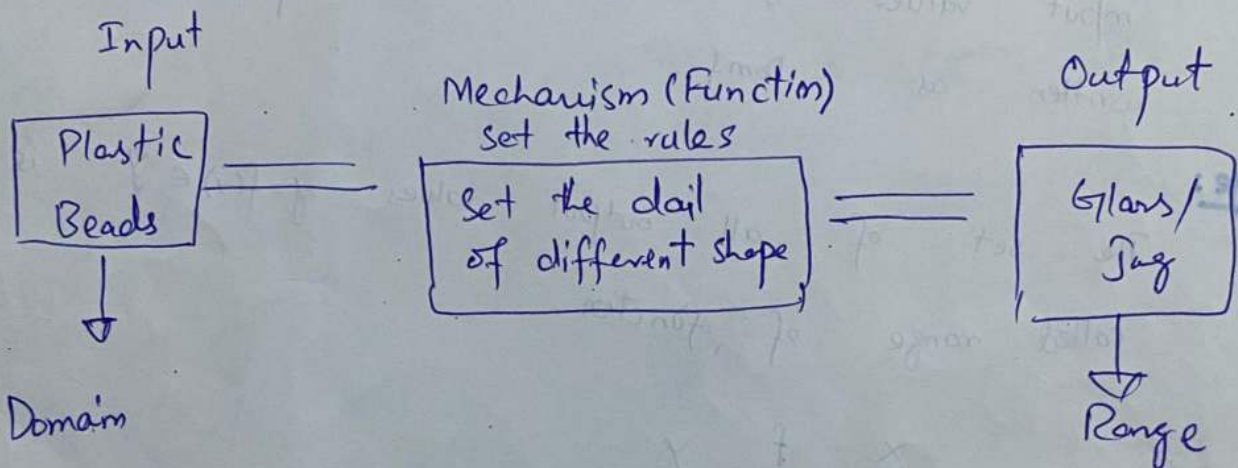
* (each element in domain) is mapped to (unique element in range)



Daily Life Example.

① Through any criteria(rule) map the students of a different group

② Plastic Machine.



Steel Beads, Glass Beads not a domain.

Function

Domain

Range

$$y = x^2$$

$$(-\infty, \infty)$$

$$[0, \infty)$$

$$y = \frac{1}{x}$$

$$\mathbb{R} \setminus \{0\}$$

from which $x \neq 0$

$$\mathbb{R} \setminus \{0\}$$

$$y = \sqrt{x}$$

$$[0, \infty)$$

value for which $x \geq 0$

$$[0, \infty)$$

$$y = \sqrt{4-x}$$

$$(-\infty, 4]$$

(value for which $4-x \geq 0$)

$$[0, \infty)$$

$$y = \sqrt{1-x^2}$$

$$[-1, 1]$$

values for which $1-x^2 \geq 0$

$$[0, 1]$$

Graph:

$$\text{Graph}(f) = \{ (x, f(x)) / x \in \text{Dom } f \} \quad \rightarrow \text{set form}$$

The graph of f is the set of all points (x, y) in cartesian plane such that x is in the domain of f and $y = f(x)$.

* To sketch the graph of $y = f(x)$, select several values of x in the domain of function $f(x)$, calculate the corresponding values of $f(x)$, plot the resulting point $(x, f(x))$ and connect the points with smooth curve.

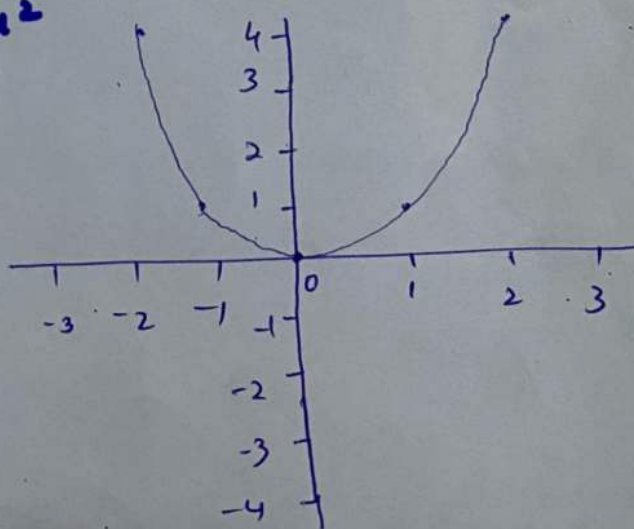
Note:

The more points you plot, the more accurate your sketch (graph) will be.

Ex: Sketch the graph of following functions

$$y = x^2$$

x	$y = f(x) = x^2$
-2	$(-2)^2 = 4$
-1	1
0	0
1	1
2	4



Linear Function:

- A function in which ① variable has power '1'.
- ② No product of variable occurs.
- ③ No trans function appear (e^x , trig, log).

→ Graph of linear function is straight line.

Vertical line test for a function.

A function can have only one value $f(x)$ for each x in its domain, so no vertical line can intersect the graph of function more than once.

Book ---

Just for understanding : Note
not for exam.

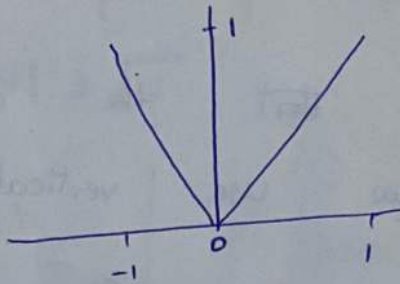
Ex: sketch the graph of $y = x^2$

x	y
-5	25
-4	16
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9
4	16
5	25

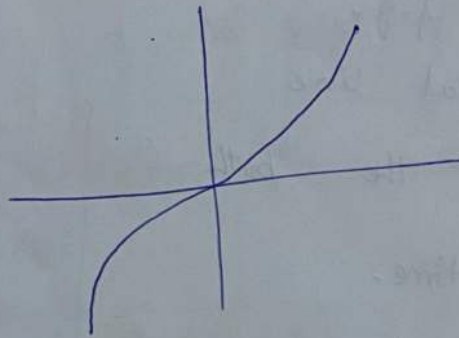
Basic functions whose graph is on fingertips

(21)

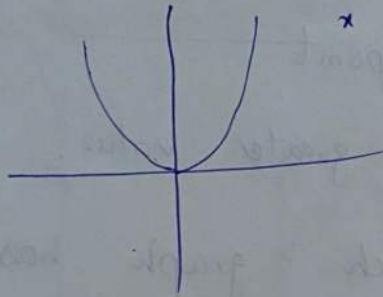
$|x|$



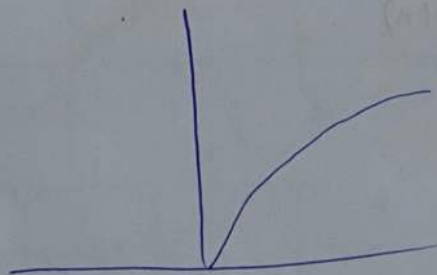
x^3



x^2



$\sqrt{x}$



Graph tells about information about function
like limit value, differentiate at point,
integration, minima maxima $\uparrow \downarrow$

In order to check ~~that~~ the comparison of two function we use vertical line.

Consider two graphs $f(x)$ & $g(x)$ as show in fig.

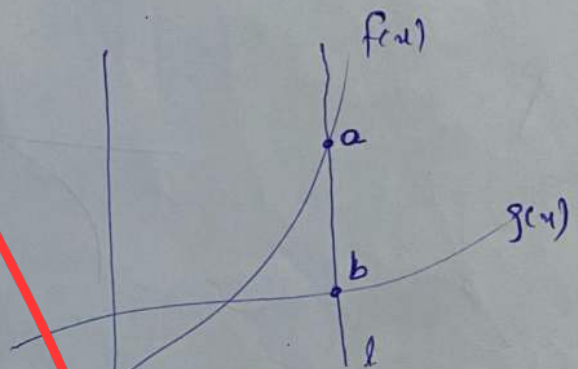
Take 'l' as vertical line

that intersect the both graphs at a time.

Now, check the ~~at~~ on

which intersecting point vertical line has greater value.

Now, check which graph has highest value of $f(x)$.



after the intersection point & point of discontinuity.

Solve the following inequality graphically.

$$|x| > \sqrt{x}$$

① Determine intersection point where

$$|x| = \sqrt{x}$$

$$S.S = \{1\}$$

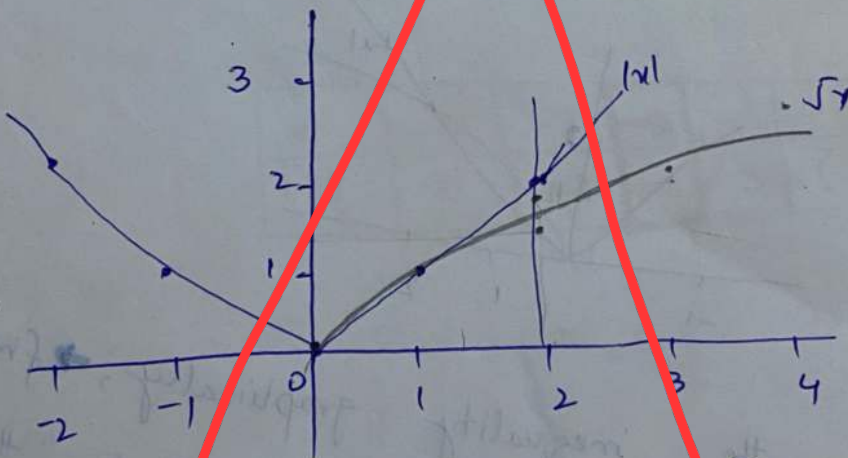
② Determine the point of discontinuity.

$$|x| = \sqrt{x}$$

No point of discontinuity.

So check the behaviour of both functions, before and after $x=1$

x	-2	-1	0	1	2
$ x $	2	1	0	1	2



x	0	1	2	3	4
$\sqrt{x}$	0	1	1.41	1.73	2

* For $x < 1$ $\sqrt{x}$ is not defined

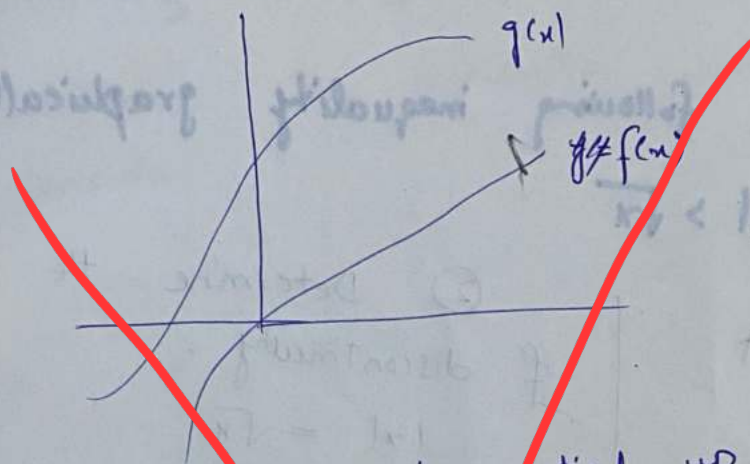
$$|x| > \sqrt{x} \rightarrow S.S = (1, \infty)$$

$$|x| \geq \sqrt{x} \rightarrow S.S = [1, \infty) \cup \{0\}$$

$$|x| < \sqrt{x} \rightarrow S.S = (0, 1)$$

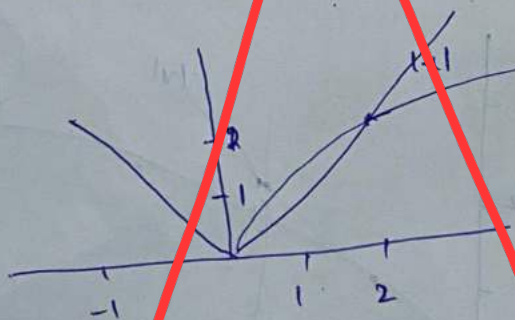
$$|x| \leq \sqrt{x} \rightarrow S.S = [0, 1]$$

Comparison of Graph:



→ The function which is vertical up has greater value than the other one.
like $g(x) > f(x)$.

(*) Some inequalities like $|x| > \sqrt{x}$, $\frac{x}{2} > 1 + \frac{4}{x}$ are not solve mathematically / analytically so we solve them graphically.



To solve the inequality graphically, → firstly ensure the intersection point to check the behaviour of function.

→ Find the undefined (point of discontinuity) to check the behaviour of function.

After intersection point and undefined (point of discontinuity) if occur, then check the behaviour before and

Solve this $\frac{x}{2} > 1 + \frac{4}{x}$ graphically.

23

① Firstly determine the intersection point. So for this consider

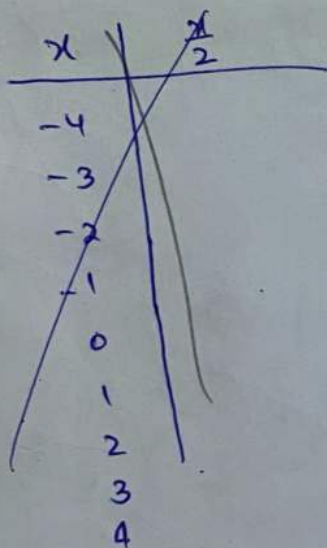
$$\frac{x}{2} = 1 + \frac{4}{x}$$

$$\Rightarrow \frac{x^2}{2} = x + 4$$

$$x^2 - 2x - 8 = 0$$

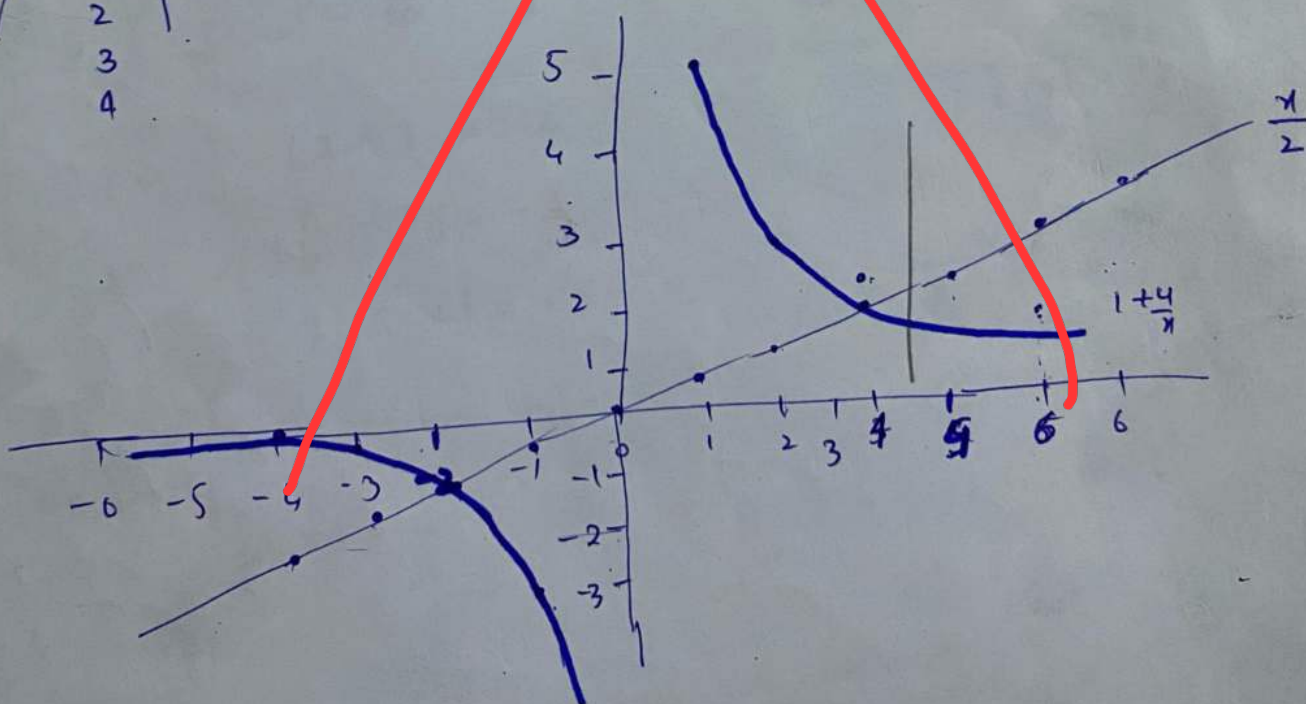
$$\Rightarrow x = -2, 4$$

② Sketch the graph before & after these points means these points are clearly visible on graph.



x	-4	-3	-2	-1	0	1	2	3	4	5	6
$\frac{x}{2}$	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3

x	-4	-3	-2	-1	0	1	2	3	4	5
$1 + \frac{4}{x}$	0	-0.33	-1	-3	?	5	3	2.33	2	1.8



Now the real line is divided b/w intersection points and undefined

So, by vertical line, we see clearly ---

$$S.S = (4, \infty) \cup (-2, 0)$$

$$\Rightarrow S.S = (-2, 0) \cup (4, \infty)$$

→ If equality i.e. $\frac{x}{2} \geq 1 + \frac{4}{x}$ so intersection points included.

$$S.S = [-2, 0) \cup [4, \infty)$$

Floor Function:

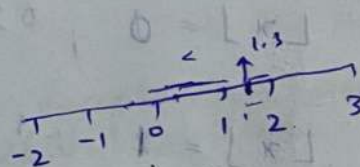
(Greatest Integer Function)

$\lfloor x \rfloor$

Def:

The function whose value at x is the greater integer less than or equal to x .

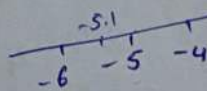
$$\lfloor 1.3 \rfloor = 1$$



$$\lfloor 2.4 \rfloor = 2$$

$$\lfloor -2.7 \rfloor = -3$$

$$\lfloor -5.1 \rfloor = -6$$



Generally

$$\lfloor x \rfloor = a$$

$$; a \leq x < a+1$$

Ceiling Function: (Least integer Function)

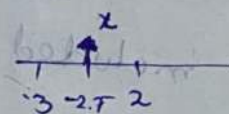
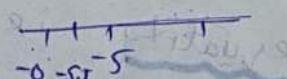
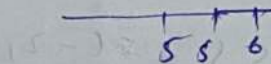
The $\lceil x \rceil$ is the smallest integer greater than or equal to x

$$\lceil 7 \rceil = 7$$

$$\lceil 5.8 \rceil = 6$$

$$\lceil -5.3 \rceil = -5$$

$$\lceil -2.7 \rceil = -2$$



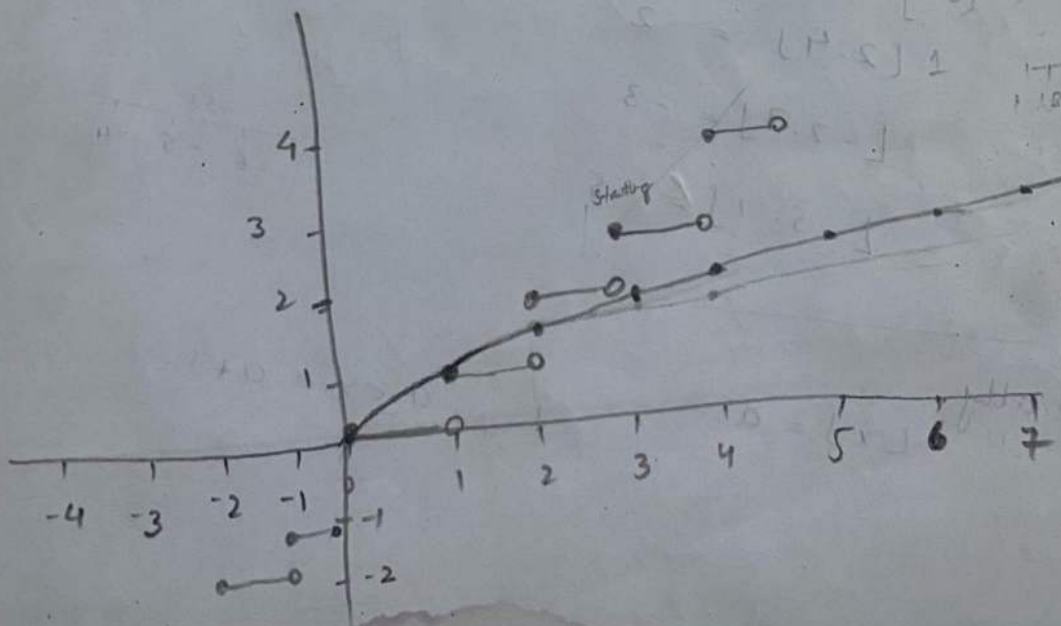
$$\lceil x \rceil = 0 ; 0 \leq x < 1$$

$$\lceil x \rceil = 1 ; 1 \leq x < 2$$

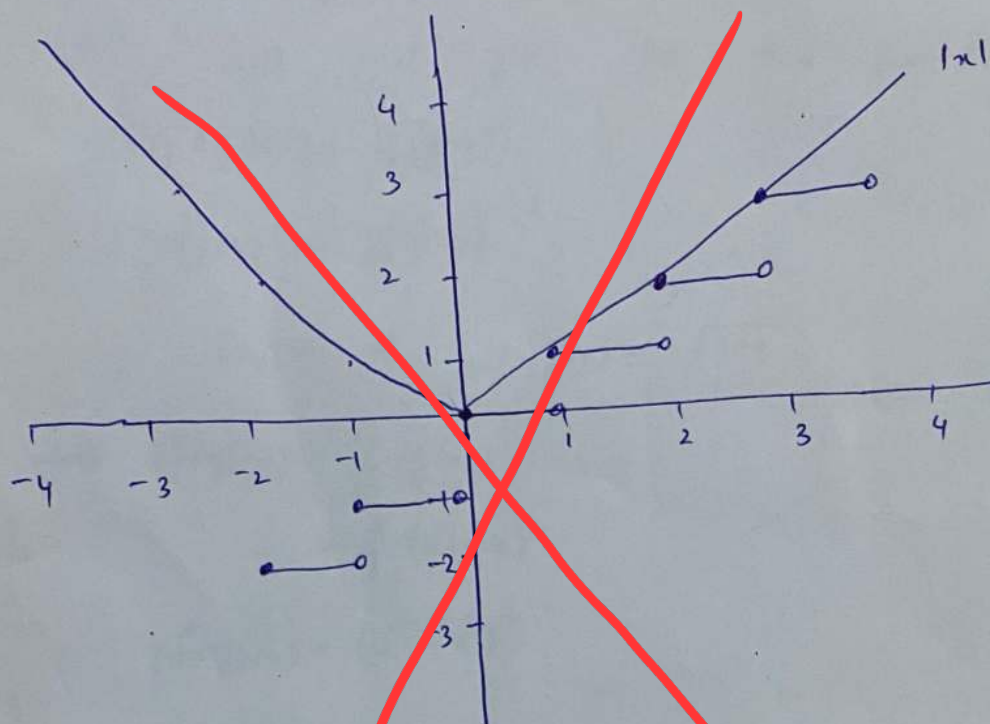
$$\lceil x \rceil = 2 ; 2 \leq x < 3$$

$$\lceil x \rceil = -1 ; -1 \leq x < 0$$

$$\lceil x \rceil = -2 ; -2 \leq x < -1$$



$$\begin{aligned} \sqrt{2} &= 1.4 \\ \sqrt{3} &= 1.73 \\ \sqrt{5} &= 2.2 \\ \sqrt{6} &= 2.4 \\ \sqrt{7} &= 2.6 \end{aligned}$$

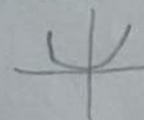


$$|x| > [x]$$

S.S = $(-\infty, 0) \cup (0, 1) \cup (1, 2) \cup (2, 3) \cup (3, 4) \dots$ All real numbers except non-negative integers.

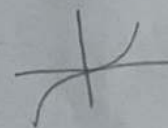
→ Even Fth: Symmetric about x-axis

$$f(x) = f(-x) \text{ e.g. } x^2$$



→ Odd Fth: Symmetric about ~~y-axis~~ ^{origin}

$$f(x) = -f(-x) \text{ e.g. } x^3$$



Algebra of Functions (Sum, Diff, Product and Quotient of function)

Let f & g be two function.

$$1) (f+g)(x) = f(x) + g(x)$$

$$2) \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

$$3) (f-g)(x) = f(x) - g(x)$$

$$4) (f \cdot g)(x) = f(x) \cdot g(x)$$

Composition of functions:

Let $f(x)$ and $g(x)$ be two functions

$$(f \circ g)(x) = f(g(x)) ;$$

$$(g \circ f)(x) = g(f(x))$$

$$f(x) = x^2 ; \quad g(x) = \sqrt{1+x}$$

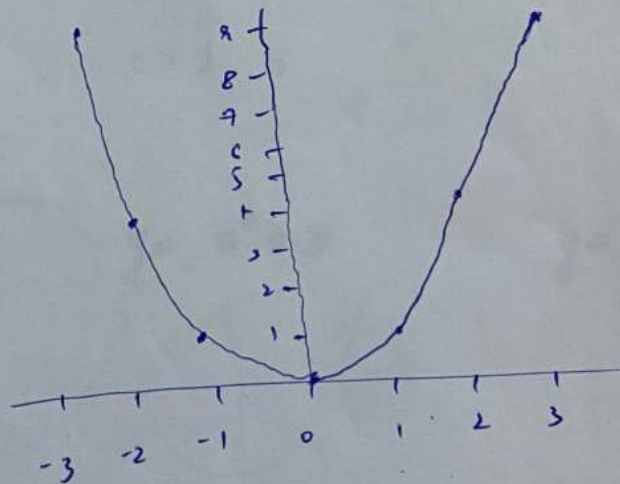
$$\begin{aligned} \rightarrow (f \circ g)(x) &= f(g(x)) \\ &= f(\sqrt{1+x}) \end{aligned}$$

$$(f \circ g)(x) = (\sqrt{1+x})^2$$

$$\begin{aligned} \rightarrow (g \circ f)(x) &= g(f(x)) \\ &= \sqrt{1+f(x)} \quad \text{or} \quad g(x^2) \\ &= \sqrt{1+x^2} \quad \text{or} \quad \sqrt{1+x^2} \end{aligned}$$

$$f(x) = x^2$$

x	x^2
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9



$\lfloor x \rfloor > \sqrt{x}$, s.s = $[2, \infty)$

$\lfloor x \rfloor \geq \sqrt{x}$, s.s = $[-1, \infty) \cup \{0, 1\}$

$\lfloor x \rfloor < \sqrt{x}$, s.s = $(0, 2) \cup (1, 2)$

$2 \leq 1.4$ $\lfloor x \rfloor \leq \sqrt{x}$, s.s = $[0, 2]$

$\lfloor 1.999 \rfloor < \sqrt{1.999}$
 $\Rightarrow 1 < 1.414$

Solve $\lfloor x \rfloor > \sqrt{x}$??



x	$\sqrt{x}$
0	0
1	1
2	1.414
3	1.732
4	2

$$y = f(x)$$

Shifting :

$$y = f(x) + k, \quad k > 0 \quad \uparrow \text{up}, \quad k < 0 \quad \downarrow \text{down (vertically)}$$

$$y = f(x + h), \quad h > 0 \quad \leftarrow \text{left}, \quad h < 0 \quad \rightarrow \text{right (horizontally)}$$

Reflection:

$$y = -f(x), \quad \text{Reflection about } x\text{-axis}$$

$$y = f(-x), \quad \text{" " } y\text{-axis}$$

Scaling :

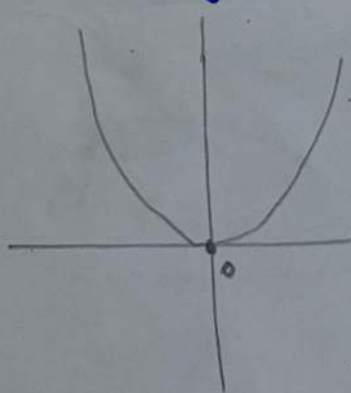
$$y = cf(x), \quad c > 1, \quad \text{vertically stretching}$$

$$y = \frac{1}{c}f(x), \quad c > 1, \quad \text{" compressing.}$$

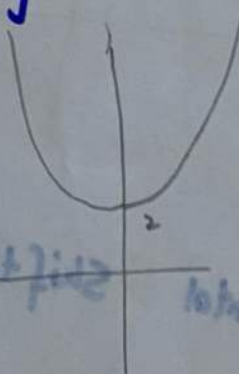
$$y = f(cx), \quad c > 1, \quad \text{horizontally}$$

$$y = f\left(\frac{x}{c}\right), \quad c > 1, \quad \text{"}$$

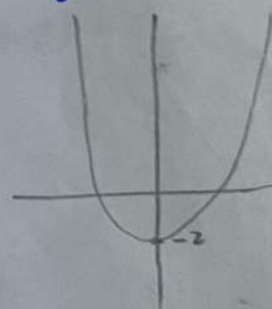
$$y = x^2$$



$$y = x^2 + 2$$



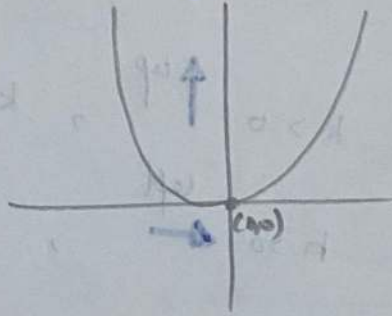
$$y = x^2 - 2$$



vertically shifting

20

$$y = x^2$$

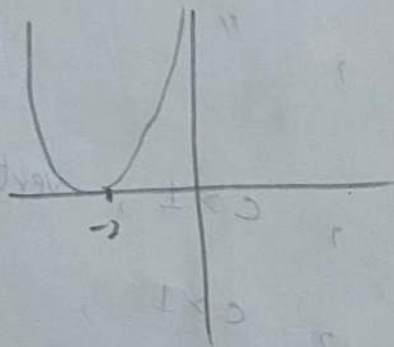


Shifting:

$$f(x) \neq k$$

$$f(x+k) = f$$

$$y = (x+2)^2$$



Reflection:

$$f(x) = -f(x)$$

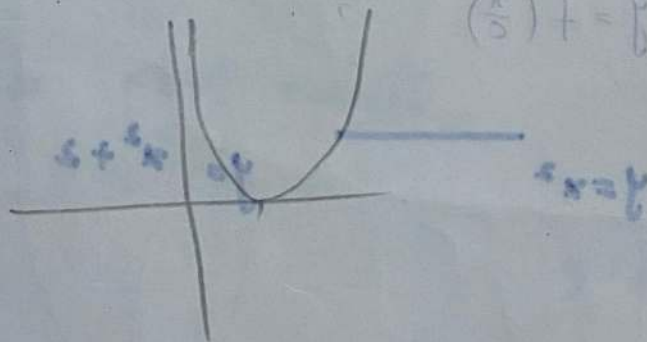
$$f(x) = f(-x)$$

Scaling:

$$f(x) = cf(x)$$

$$f(x) = \frac{1}{c}f(x)$$

$$y = (x-2)^2$$



$$f(x) = f(x)$$

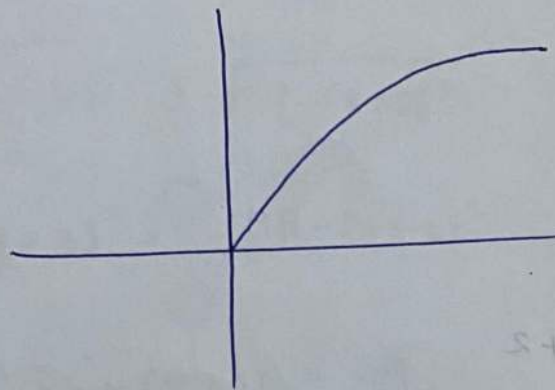
$$f\left(\frac{x}{c}\right) = f$$

Horizontal Shifting

vertically shifting

$$y = \sqrt{x}$$

$$y = f(x)$$

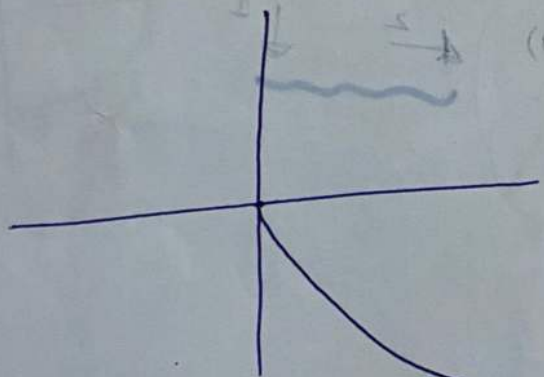


$$2 + \sqrt{x} = f(x)$$

$$x + (2 - \sqrt{x}) = 0$$

$$y = -\sqrt{x}$$

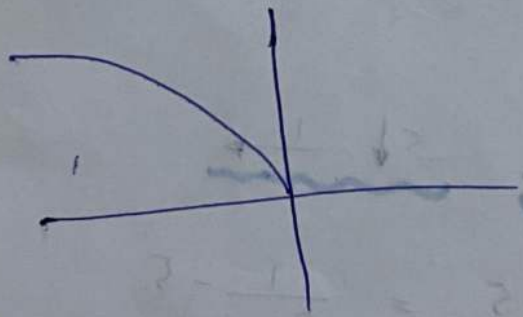
$$y = -f(x)$$



Reflection about x-axis.

$$y = \sqrt{-x}$$

$$y = f(-x)$$



Reflection about y-axis

1- $y = x^2 + 5 = f(x)$; 2[↑] 3[→]

For 3 $\rightarrow$

$$y = f(x-3)$$

Then for 2[↑]

$$y = f(x-3) + 2$$

$$= ((x-2)^3 + 5) + 2$$

$$\therefore f(x) = x^3 + 5$$

$$y = (x-2)^3 + 7$$

2- $y = \frac{1}{x^2} = f(x)$ 2[←] 1[↓]

For 2[←]

$$y = f(x+2)$$

For 1[↓]

$$y = f(x+2) - 1$$

$$y = \frac{1}{(x+2)^2} - 1$$

3- $y = \frac{1}{8(x+1)}$; 5[↓] 1[→]

$$y = \frac{1}{8(x+1-1)} - 5 = \frac{1}{8x} - 5$$

$$y = 10x - 7$$

28

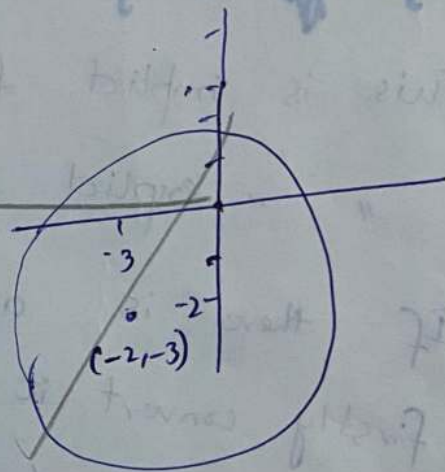
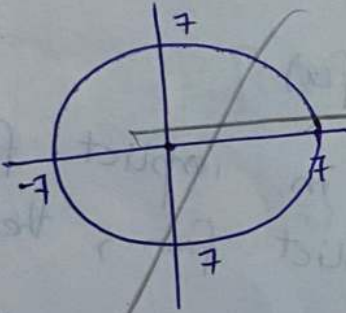
$$y = 10x - 7$$

$$y + 3 = \pm \sqrt{49 - (x+2)^2}$$

$$(y+3)^2 = 49 - (x+2)^2$$

$$(x+2)^2 + (y+3)^2 = 49$$

Given



4- $y = 2x - 7$; $7 \uparrow$

128

$$y = f(x) = 2x - 7$$

For vertically upward $y = f(x) + k$

$$y = 2x - 7 + 7$$

$$y = 2x$$

5- $x^2 + y^2 = 49$ (Explicit form of function) $3\frac{1}{2}, 4^2$

This is implicit form $f(x, y) = c$ of function

" " explicit form $y = f(x)$

If there is a function in implicit form, firstly convert it into explicit form, then perform the operations according to scaling, shifting whatever. After that, convert it into original form (implicit form).

So, $x^2 + y^2 = 49$ is in implicit form i.e. $f(x, y) = c$

so (convert it in explicit form i.e. $y = f(x)$)

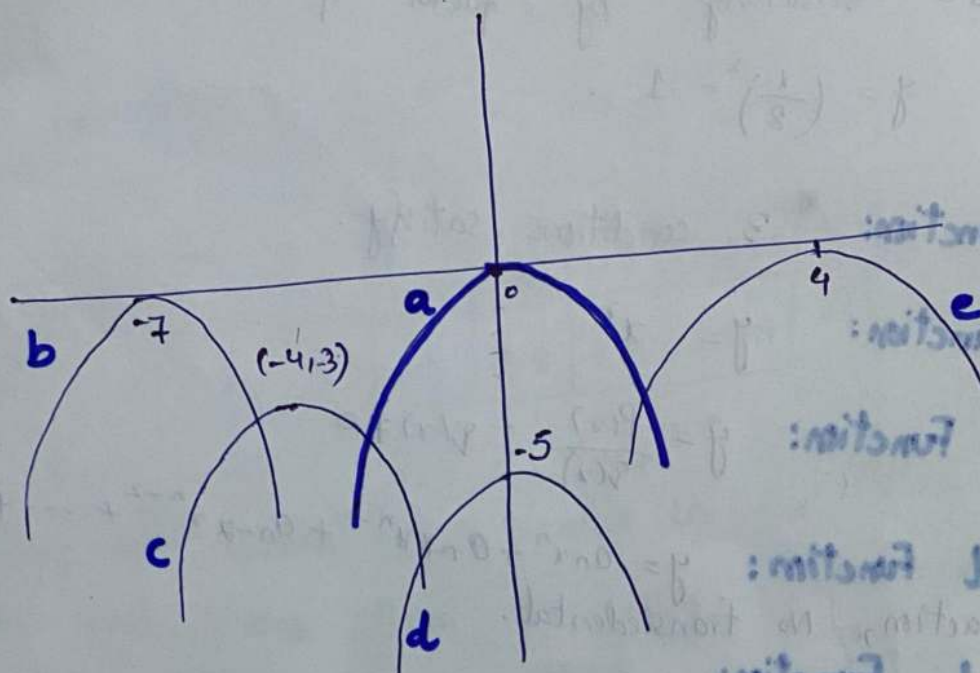
$$\Rightarrow y^2 = 49 - x^2$$

$$y = \pm \sqrt{49 - x^2} = f(x)$$

Now,

$$y = \pm \sqrt{49 - (x+2)^2} - 3$$

or Now convert it in given form i.e. implicit form



a: $f(x) = -x^2$

b: $f(x) = -(x+7)^2$

c: $f(x) = -(x+4)^2 - 3$

d: $f(x) = -x^2 - 5$

e: $f(x) = -(x-4)^2$

Ex related to Scaling.

$y = x^2 - 1$

→ Stretching vertically by factor of 3.
 $y = 3(x^2 - 1)$

→ compression vertically by factor of 3.
 $y = \frac{1}{3}(x^2 - 1)$

→ Horizontal compression by factor of 5.
 $y = (5x)^2 - 1$

PS

→ Horizontal stretching by factor of 8.

$$y = \left(\frac{x}{8}\right)^2 - 1$$

Linear Function: 3 conditions satisfy.

Power Function: $y = x^n$

Fractional Function: $y = \frac{P(x)}{Q(x)}$; $Q(x) \neq 0$

Polynomial Function: $y = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0$
No fraction, No transcendental.

Exponential Function:

when variable is as exponent (power) and constant as a base.

Trigonometric Function:

$\sin x$, $\cos x$, $\tan x$ and related trigonometric function.

Logarithmic Function:

$$\log x$$

Ex related to scaling.

$$y = 4x - 1$$

$$y = 3(1 - 1)$$

$$y = \frac{1}{2}(1 - 1)$$

$$y = 1 - 1$$

→ Limit:

$$y = f(x) = \frac{x^2 - 4}{x - 2}, \quad x \neq 2$$

x	-1	0	1	1.3	1.8	1.9	1.9999	2	2.001	2.01	2.5	3	3.999	3
$y = f(x)$	1	2	3	3.3	3.8	3.9	3.9999	?	4.001	4.2	4.5	4.8	4.999	5

← left

right →

every real number is located in real line.

In real line there is only two direction left & right to reach a point.

So from above table it is clearly seen

that when we move towards 2 from right

$$x: 3 \rightarrow 2$$

$$y = f(x): 5 \rightarrow 4$$

Also, when we move towards 2 from left

$$x: 1 \rightarrow 2$$

$$y = f(x): 3 \rightarrow 4$$

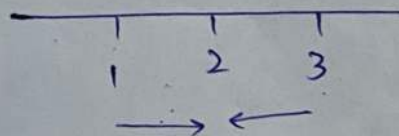
For $x \rightarrow a$.

$f(x) \rightarrow l$ certain number

we say limit of function exist.

$$\text{i.e. } \lim_{x \rightarrow a} f(x) = l$$

read it as, when x approach to a ,
 $f(x)$ " " l .



At $x=2$, function has no value at that point. (31)

But $x \rightarrow 2$.

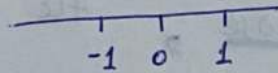
$$f(x) = \frac{x^2 - 4}{x - 2} \rightarrow 4.$$

In order to check limit exist.

We check,

$$L.H.L = R.H.L$$

Q) $\lim_{x \rightarrow 0} \sqrt{x}$



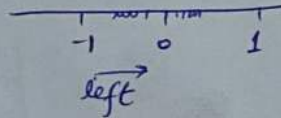
$$L.H.L = \lim_{x \rightarrow 0^-} \sqrt{x} \neq \text{any real number.}$$

$$R.H.L = \lim_{x \rightarrow 0^+} \sqrt{x} = \sqrt{0.00001} \approx 0.$$

This implies L.H.L does not exist.

Therefore $\lim_{x \rightarrow 0} \sqrt{x}$ does not exist.

Q) $\lim_{x \rightarrow 0} [x]$



$$L.H.L = \lim_{x \rightarrow 0^-} [x]$$

$$= \lim_{x \rightarrow 0^-} [-0.000001] = -1.$$

$$R.H.L = \lim_{x \rightarrow 0^+} [x] = [0.000001] = 0.$$

Since, $L.H.L \neq R.H.L$

So, $\lim_{x \rightarrow 0} [x]$ does not exist.

007
004
052
008
055
010

Remark:

In case, where function is discontinuous at that point, there is chance of that limit is not equal, so limit does not exist.

Now query arise here that, at every point,
we check L.H.L & R.H.L.

It is not necessary to check L.H.L & R.H.L

at every point, if ~~at~~ of domain.

At where check L.H.L & R.H.L: Floor Fwd ...

But if 1) function is piecewise, the point where
its domain is break. At that pt we check

L.H.L & R.H.L

Also at 2) the boundary of domain, at the

3) point of discontinuity (where the function)
is undefined (not-defined).

Remark:

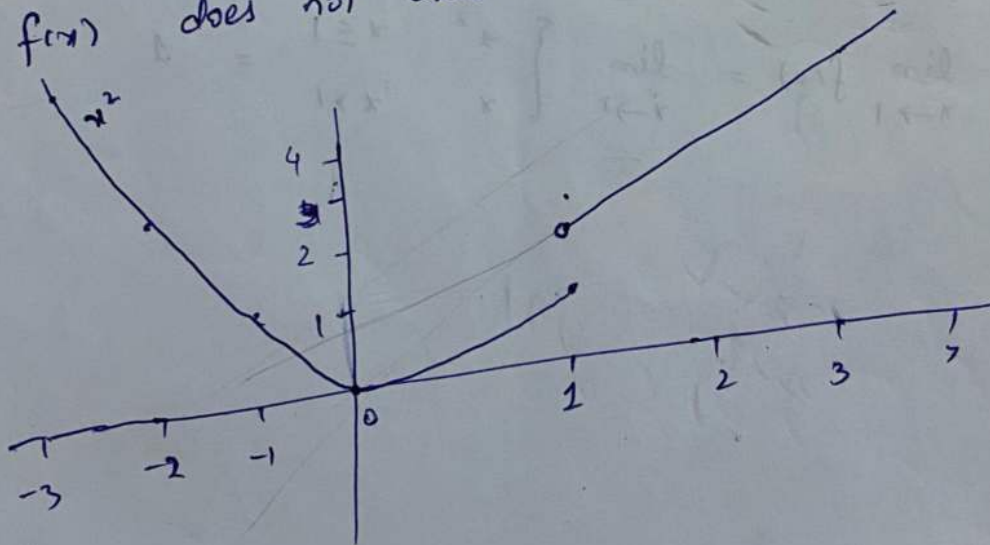
$$f(x) = \begin{cases} x^2 & x \leq 1 \\ x+1 & x \geq 1 \end{cases}$$

L.H.L : $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2$
 $= 1$

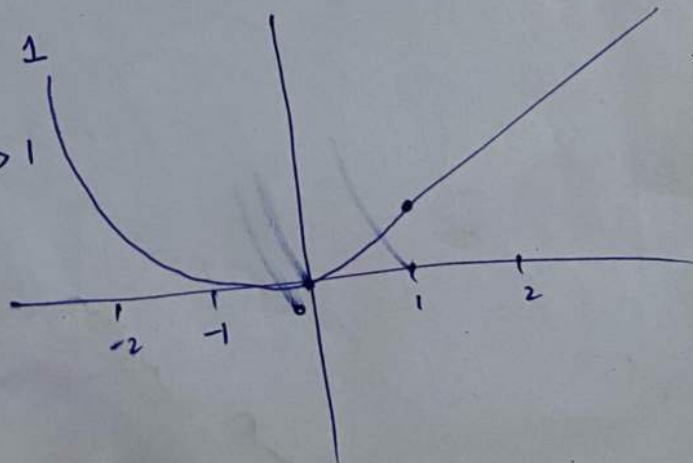
R.H.L : $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (x+1)$
 $= 2$

$L.H.L \neq R.H.L$

$\lim_{x \rightarrow 1} f(x)$ does not exist.



$$f(x) = \begin{cases} x^2 & x \leq 1 \\ x & x > 1 \end{cases}$$



L.H.L

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1$$

R.H.L

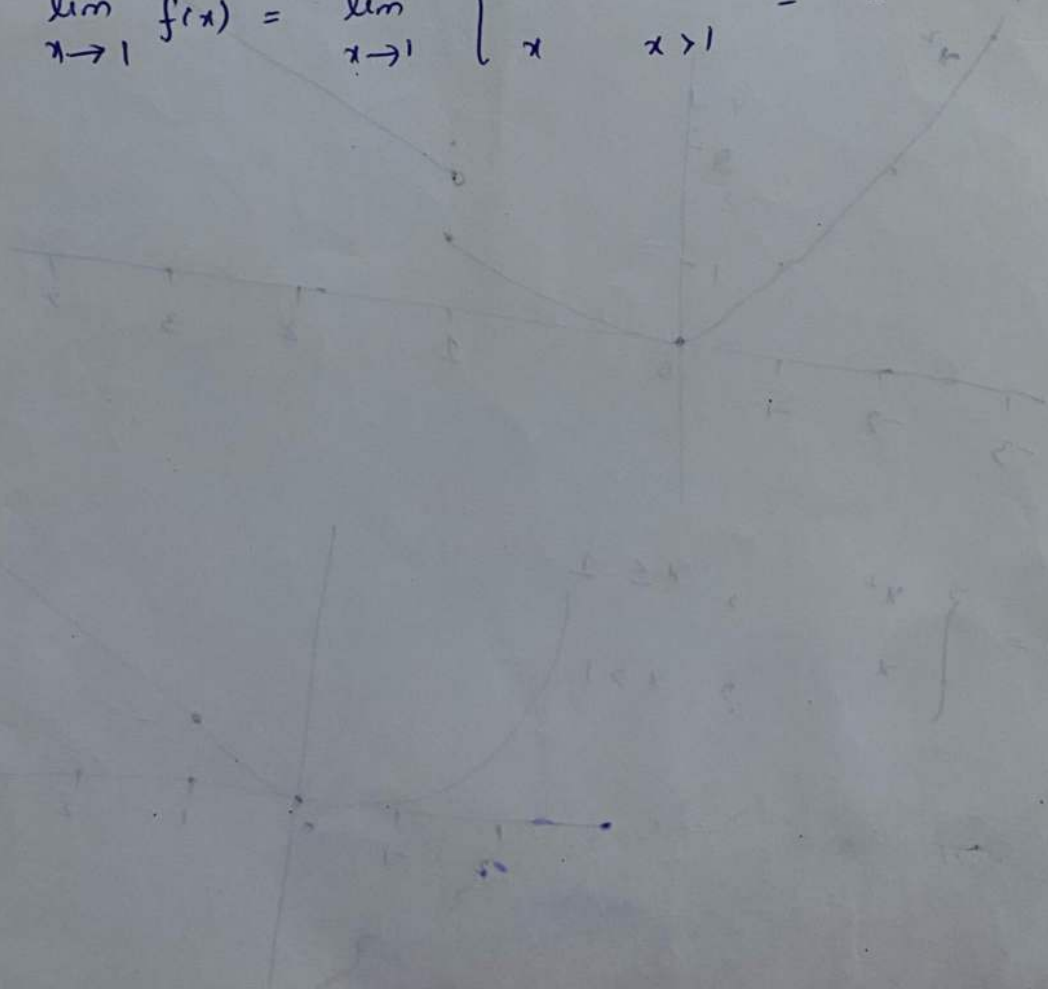
$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} x = 1$$

* Now here, you put directly $x = 1$ in $f(x)$ rather than any point nbh. in 1.

$$L.H.L = R.H.L = 1$$

So, the limit exist. that is 1

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \begin{cases} x^2 & x \leq 1 \\ x & x > 1 \end{cases} = 1$$



Q1) $\lim_{x \rightarrow 0} \frac{1}{x}$?

We check both L.H.L & R.H.L b/c Domain of $\frac{1}{x}$ is $[0, \infty)$

L.H.L

$$\lim_{x \rightarrow 0^-} \frac{1}{x} = \frac{1}{-0.000001}$$

$$= -\infty$$

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \frac{1}{0.000001} = \infty$$

L.H.L is not defined.

Solve $\lim_{x \rightarrow 1} \lfloor x - 1 \rfloor = ?$



$$\lim_{x \rightarrow 1^-} \lfloor 0.9999 - 1 \rfloor = \lfloor -0.0001 \rfloor = -1$$

$$\lim_{x \rightarrow 1^+} \lfloor 1.0001 - 1 \rfloor = \lfloor 0.0001 \rfloor = 0$$

$$L.H.L \neq R.H.L$$

$$\rightarrow f(x) \quad [2, \infty)$$

Check $\lim_{x \rightarrow 2}$ without evaluating limit does not exist at $x=2$
b/c L.H.L

$$\rightarrow f(x) \quad [-\infty, 3]$$

$\lim_{x \rightarrow 3}$ without evaluating limit does not exist at $x=3$
b/c R.H.L

$\rightarrow f(x) \quad (-\infty, 1) \cup (1, \infty)$ limit may exist or not.

e.g $f(x) = \lim_{x \rightarrow 1} \frac{x}{x-1}$

$$\underline{L.H.L} \quad \lim_{x \rightarrow 1^-} \frac{x}{x-1} = \frac{0.9999}{0.9999-1} = \frac{0.9999}{-0.0001} = -\infty$$

$$\lim_{x \rightarrow 1^+} \frac{x}{x-1} = \infty$$

Properties of Limit:

Suppose $\lim_{x \rightarrow a} f(x) = L$ &

$$\lim_{x \rightarrow a} g(x) = M.$$

135

$$1) \lim_{x \rightarrow a} c = c$$

$$2) \lim_{x \rightarrow a} x = a.$$

$$3) \lim_{x \rightarrow a} (f(x) \pm g(x)) = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = L \pm M.$$

$$4) \lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = L \cdot M.$$

$$5) \lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \neq \frac{L}{M}, M \neq 0.$$

6) If $f(x) = g(x)$ for all x except possibly at $x = a$.

$$\text{then } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x).$$

$$\text{e.g. } \frac{x^2 - 1}{x - 1} = x + 1$$

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} x + 1$$

$$6) \lim_{x \rightarrow a} (c f(x)) = c \lim_{x \rightarrow a} f(x)$$

$$7) \lim_{x \rightarrow a} (f(x))^{r/s} = \left(\lim_{x \rightarrow a} f(x) \right)^{r/s} = L^{r/s}$$

where r and s has no common divisor except 1 and

$$L > 0.$$

e.g:

$$\begin{aligned}\lim_{x \rightarrow 2} \sqrt{x^2 - 1} &= \lim_{x \rightarrow 2} (x^2 - 1)^{1/2} \\ &= \sqrt{\lim_{x \rightarrow 2} (x^2 - 1)} \\ &= \sqrt{(2)^2 - 1} \\ &= \sqrt{3}\end{aligned}$$

$$\lim_{x \rightarrow 0} \sqrt{x} = \left(\lim_{x \rightarrow 0} x \right)^{1/2}$$

$$X = (0)^{1/2} = 0$$

Here $f(x) = x$

$$\lim_{x \rightarrow 0} f(x) = L = 0 \text{ but } L > 0.$$

$$\textcircled{8} \quad \lim_{x \rightarrow a} |f(x)| = \left| \lim_{x \rightarrow a} f(x) \right| = |L|.$$

(Independence Property)

$\textcircled{9}$ If $f(x) \leq g(x)$ for all values of x except possibly at $x = a$.

$$\text{then } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x).$$

$\therefore$ b/c limit is functional value at nbhd of a .

$$\text{e.g. } \frac{x^2 - 1}{x - 1} = x + 1$$

$$\lim_{x \rightarrow 1} () = \lim (x + 1)$$

$$2 = 2.$$

$\textcircled{10}$ If $f(x) \leq g(x)$ for all x except possibly at $x = a$.

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &\leq \lim_{x \rightarrow a} g(x) \\ \Rightarrow L &\leq M\end{aligned}$$

Squeezing Property / Sandwich Theorem.

136

(11) If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = L$

and $f(x) \leq h(x) \leq g(x)$

$$\Rightarrow \lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} h(x) \leq \lim_{x \rightarrow a} g(x)$$

$$\Rightarrow \lim_{x \rightarrow a} h(x) = L$$

(12) If $P(x)$ be a polynomial.

i.e. $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

(No fraction, No power in radical/fraction)

Domain = $\mathbb{R}$.

then $\lim_{x \rightarrow a} P(x) = P(a)$.

(13) If $f(x) = \frac{P(x)}{Q(x)}$; $Q(x) \neq 0$

then $\lim_{x \rightarrow a} f(x) = \frac{\lim_{x \rightarrow a} P(x)}{\lim_{x \rightarrow a} Q(x)} = \frac{P(a)}{Q(a)}$; $Q(a) \neq 0$.

e.g. $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 1}{x + 5} = \frac{\lim_{x \rightarrow 1} x^2 + 2x - 1}{\lim_{x \rightarrow 1} x + 5}$

$$= \frac{(1)^2 + 2(1) - 1}{1 + 5}$$

$$= \frac{2}{6} = \frac{1}{3}$$

$$(14) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$(15) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

EX NO 1.7

Numerically Method
to solve Limit

Q1 - 14.

EX NO 1.8

Q1 - 66

Ex Q: Randomly

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x} = \frac{\sin \pi}{\pi} = \frac{0}{\pi} = 0.$$

$$Q \lim_{x \rightarrow 0} \frac{x}{\sin 3x}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{3x}{3 \sin 3x}$$

$$\lim_{x \rightarrow 0} \frac{1}{3 \left(\frac{\sin 3x}{3x} \right)}$$

$$= \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$$

$$= \frac{1}{3} \lim_{3x \rightarrow 0} \frac{\sin 3x}{3x}$$

$$= \frac{1}{3} \cdot 1$$

$$= \frac{1}{3}$$

$$\begin{matrix} x \rightarrow 0 \\ 3x \rightarrow 0 \end{matrix}$$

Q) $\lim_{x \rightarrow 0} \frac{|x|}{3 - \sqrt{9+x}}$

Sol:

$$\therefore |x| = \begin{cases} x & , x \geq 0 \\ -x & , x < 0 \end{cases}$$

$$f(x) = \begin{cases} \frac{x}{3 - \sqrt{9+x}} & ; x \geq 0 \\ \frac{-x}{3 - \sqrt{9-x}} & ; x < 0 \end{cases}$$

Since the function is break at $x=0$. So we compute L.H.L and R.H.L at '0'.

L.H.L:

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{-x}{3 - \sqrt{9-x}}$$

$$= \lim_{x \rightarrow 0^-} \frac{-0}{3 - \sqrt{9-x}} \quad (\text{undefined}) \quad (\text{indeterminant form})$$

So we rationalize.

$$= \lim_{x \rightarrow 0^-} \frac{-x}{3 - \sqrt{9-x}} \times \frac{3 + \sqrt{9-x}}{3 + \sqrt{9-x}}$$

$$= \lim_{x \rightarrow 0^-} \frac{-x(3 + \sqrt{9-x})}{(3)^2 - (\sqrt{9-x})^2}$$

$$= \lim_{x \rightarrow 0^-} \frac{-x(3 + \sqrt{9-x})}{9 - (9-x)} = \frac{-x(3 + \sqrt{9-x})}{a - 9 + x}$$

$$= \lim_{x \rightarrow 0^-} \frac{-x(3 + \sqrt{9-x})}{x}$$

$$= \lim_{x \rightarrow 0^-} -(3 + \sqrt{9-x})$$

$$= -(3 + \sqrt{9-0})$$

$$= -(3+3)$$

$$= -6$$

R.H.L:

$$= \lim_{x \rightarrow 0^+} \frac{x}{3 - \sqrt{9+x}}$$

By applying limit directly we get indetermined form.

So rationalize it.

$$= \lim_{x \rightarrow 0^+} \frac{x}{3 - \sqrt{9+x}} \times \frac{3 + \sqrt{9+x}}{3 + \sqrt{9+x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{x(3 + \sqrt{9+x})}{(3)^2 - (\sqrt{9+x})^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{x(3 + \sqrt{9+x})}{9 - (9+x)}$$

$$= \lim_{x \rightarrow 0^+} \frac{x(3 + \sqrt{9+x})}{-x}$$

$$= -(3 + \sqrt{9+x})$$

$$= -(3 + \sqrt{9})$$

$$= -6$$

Since L.H.L = R.H.L

So the limit exist.

$$\text{i.e. } \lim_{x \rightarrow 0} \frac{|x|}{3 - \sqrt{9+|x|}} = -6.$$

$$Q) \lim_{x \rightarrow \pi} \frac{1 + \cos x}{(\pi - x)^2}$$

$$= \lim_{x \rightarrow \pi} \frac{1 + \cos x}{(\pi - x)^2} \times \frac{1 - \cos x}{1 - \cos x}$$

$$= \lim_{x \rightarrow \pi} \frac{1 - \cos^2 x}{(\pi - x)^2 (1 - \cos x)}$$

$$\therefore \sin^2 x + \cos^2 x = 1$$

$$= \lim_{x \rightarrow \pi} \frac{\sin^2 x}{(\pi - x)^2 (1 - \cos x)}$$

$$\therefore \sin(\pi - x) = \sin x$$

$$= \lim_{x \rightarrow \pi} \frac{\sin^2(\pi - x)}{(\pi - x)^2 (1 - \cos x)}$$

$$= \lim_{x \rightarrow \pi} \left(\frac{\sin(\pi - x)}{(\pi - x)} \right)^2 \cdot \frac{1}{1 - \cos x}$$

$$= \lim_{x \rightarrow \pi} \left(\right) \cdot \lim_{x \rightarrow \pi} \frac{1}{1 - \cos x}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$= 1 \cdot \frac{1}{1 - \cos \pi}$$

$$= 1 \cdot \frac{1}{1 - (-1)}$$

$$= \frac{1}{2}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\therefore \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$\cos(\pi - x) = -\cos x$$

$$\frac{1 - \cos(\pi - x)}{\pi - x} \cdot \frac{1}{\pi - x}$$

= 0

No approachable

$$\sin(\pi - x)$$

$$\sin \pi \cos x - \cos \pi \sin x$$

$$= -(-\sin x) = \sin x$$

$$\therefore \cos \pi = -1$$

$$Q) \lim_{t \rightarrow 0} \frac{\tan 4t}{2t} \quad \left(\frac{0}{0}\right)$$

$$\text{Sol:} \\ = \lim_{t \rightarrow 0} \frac{\frac{\sin 4t}{\cos 4t}}{2t}$$

$$= \lim_{t \rightarrow 0} \frac{\sin 4t}{2t(\cos 4t)}$$

$$= \lim_{t \rightarrow 0} \frac{\sin 4t}{2t} \cdot \frac{1}{\cos 4t}$$

$$= \lim_{t \rightarrow 0} \frac{2 \sin 4t}{4t} \cdot \frac{1}{\cos 4t}$$

$$\therefore \begin{aligned} t &\rightarrow 0 \\ 4t &\rightarrow 0.4 \\ 4t &\rightarrow 0 \end{aligned}$$

$$= 2 \lim_{4t \rightarrow 0} \frac{\sin 4t}{4t} \cdot \lim_{4t \rightarrow 0} \frac{1}{\cos 4t}$$

$$= 2(1) \cdot$$

$$= 2$$

$$Q) \lim_{t \rightarrow 0} \frac{1 - \cos^2 t}{t^2}$$

$$\therefore \sin^2 t + \cos^2 t = 1$$

$$\Rightarrow \sin^2 t = 1 - \cos^2 t$$

$$= \lim_{t \rightarrow 0} \frac{\sin^2 t}{t^2}$$

$$= \lim_{t \rightarrow 0} \left(\frac{\sin t}{t} \right)^2$$

$$= \left(\lim_{t \rightarrow 0} \frac{\sin t}{t} \right)^2 = (1)^2 = 1$$

Continuity at a point:

139

A function f is continuous at a number a if following conditions holds

- (i) $f(a)$ is defined.
- (ii) $\lim_{x \rightarrow a} f(x)$ exists $\Leftrightarrow$ L.H.L = R.H.L
- (iii) $\lim_{x \rightarrow a} f(x) = f(a)$.

$$\Rightarrow \text{L.H.L} = \text{R.H.L} = f(a)$$

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a).$$

Exp: $f(x) = |x|$

$$|x| = \begin{cases} x & , x \geq 0 \\ -x & , x < 0 \end{cases}$$

L.H.L :

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -x = 0.$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0.$$

$$\therefore \text{L.H.L} = \text{R.H.L}$$

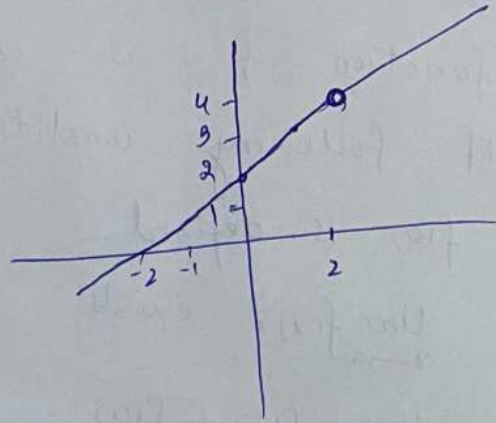
$\Rightarrow$ limit exist.

$$f(0) = 0$$

$$\Rightarrow \text{L.H.L} = \text{R.H.L} = f(0)$$

$\Rightarrow$ $|x|$ function is continuous at $x=0$.

Q) $f(x) = \frac{x^2 - 4}{x - 2}$



→ Mathematically:

$\lim_{x \rightarrow 2} f(x) = 6$ (From previous lec)

$f(2)$ is undefined.

The 1st condition of continuity at a point violates.

Hence, $f(x)$ is not cts at $x=2$.

→ Graphically,

The graph of given function i.e. $f(x) = \frac{x^2 - 4}{x - 2}$, has hole (a break).

so, $f(x)$ is not a continuous.

Q $f(x) = \frac{|x|}{x}$

check the continuity at $x=0$.

Sol:

~~for~~ $|x| = \begin{cases} x & , x \geq 0 \\ -x & , x < 0 \end{cases}$

$\Rightarrow \frac{|x|}{x} = \begin{cases} 1 & , x \geq 0 \\ -1 & , x < 0 \end{cases}$

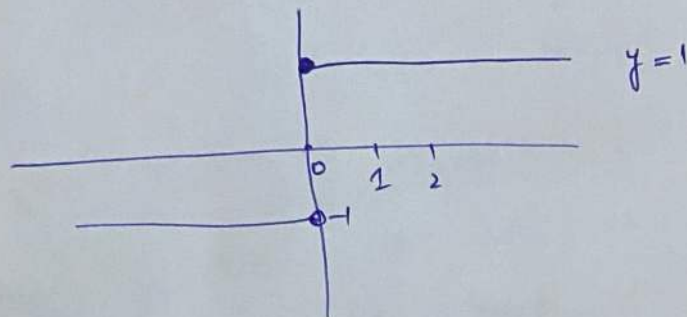
L.H.L $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -1 = -1$

R.H.L $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 1 = 1$

$$\therefore L.H.L \neq R.H.L$$

$\Rightarrow$ limit does not exist.

$\Rightarrow f$ is not cts.



$\Rightarrow$ Graph of $\frac{|x|}{x}$ show gap, hence it is not continuous ftn.

Remark;

① If the graph of function, contain a gap of one point like $f(x) = \frac{x^2 - 4}{x - 2}$, such type of discontinuity called point discontinuity (in case of point discontinuity, $f(a)$ is undefined).

② If the graph of function, contain a gap of more than one point like $f(x) = \frac{|x|}{x}$, such type of discontinuity called jump discontinuity. (In this case, limit does not exist)

How to check continuity of function on a interval?

we read about the continuity at a certain point. An interval contain infinite real number, so continuity at a point collapse on a interval.

→ we find point of discontinuity. means the point where function become undefined.

- ✓ In case of rational ftn these where $den = 0$.
- ✓ " " " square root ftn where $\sqrt{} < 0$.
- ✓ " " " piecewise ftn may or may not where domain break.

Ex NO 1.9 Q(35-43)

Q39- $G(x) = \frac{4x-3}{16x^2-9}$, $[-\frac{3}{4}, 0]$, $[-\frac{1}{2}, 0]$, $(-\frac{3}{4}, \infty)$, $[-2, \infty]$ & $(-1, -\frac{3}{4})$.

Sol: We check ftn is continuous for ~~at~~ ^{discont} on interval.

So, we find, point of discontinuity.

Given ftn is rational ftn

consider, denominator = $16x^2 - 9 = 0$
 $\Rightarrow x^2 = \frac{9}{16}$

or $x = \pm \frac{3}{4}$.

$\Rightarrow$ At $x = -\frac{3}{4}, +\frac{3}{4}$, $G(x)$ is undefined (discontinuous).

$\Rightarrow -\frac{3}{4}, +\frac{3}{4}$ are point of discontinuities.

located these pt of " " in given interval.

$$\pm \frac{3}{4} \in [-\frac{3}{4}, 0], (-\frac{3}{4}, \infty), [-2, \infty)$$

$\Rightarrow f(x)$ is discontinuous at on these above mentioned intervals.

$$\Rightarrow \pm \frac{3}{4} \notin [-\frac{1}{2}, 0], (-1, -\frac{3}{4})$$

$\Rightarrow u(x)$ is continuous on $[-\frac{1}{2}, 0], (-1, -\frac{3}{4})$

Q43 - we find 'a' so that g will be continuous at 1.

Since $g(x)$ is continuous at 1

$$L.H.L = R.H.L = f(1)$$

$$\Rightarrow \lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^+} g(x) = g(1) \quad (*)$$

$$\text{i.e. } \lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} \frac{x^3 - 1}{x - 1} = \frac{(x-1)(x^2 + x + 1)}{(x-1)}$$

$$\lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} \frac{x^3 - 1}{x - 1} = \frac{(x-1)(x^2 + x + 1)}{(x-1)}$$

Q43-

Since $g(x)$ is continuous at $x=1$

142

$$\Rightarrow \lim_{x \rightarrow 1} g(x) = g(1) \quad (*)$$

$$\begin{aligned} \lim_{x \rightarrow 1} g(x) &= \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{(x-1)(x^2 + x + 1)}{(x-1)} \\ &= \lim_{x \rightarrow 1} (x^2 + x + 1) \\ &= (1)^2 + (1) + 1 \end{aligned}$$

$$\lim_{x \rightarrow 1} g(x) = 3$$

$$g(1) = a.$$

$$(*) \Rightarrow \boxed{a=3}$$

$$Q - f(x) = \begin{cases} x^2 & , x \leq 1 \\ x & , x > 1 \end{cases}$$

Mathematically:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1$$

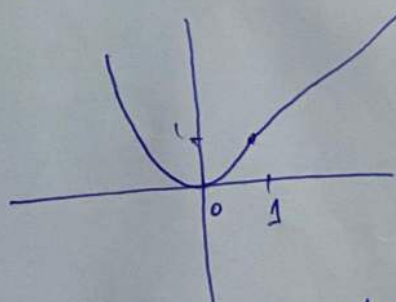
$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} x = 1$$

$$f(1) = (1)^2 = 1$$

$$\Rightarrow L.H.L = R.H.L = f(1)$$

$\Rightarrow f(x)$ is cts at $x=1$

Graphically:



Graph has no hole/break,
So the $f(x)$ is cts.

54

$$f(x) = \begin{cases} 3+x, & x \leq 1 \\ 3-x, & x > 1 \end{cases}$$

Mathematically,

L.H.L

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1} (3+x) \\ &= 4 \end{aligned}$$

R.H.L

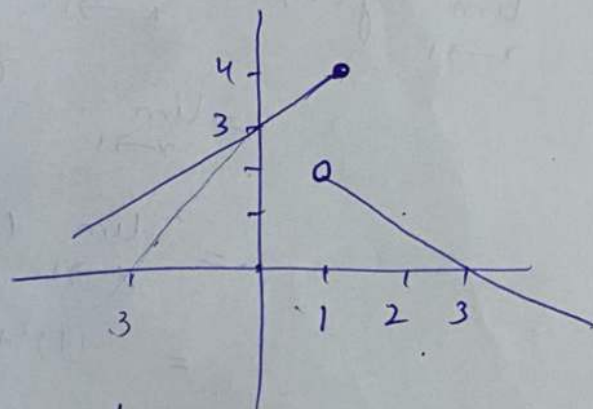
$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1} (3-x) \\ &= 3-1 \\ &= 2 \end{aligned}$$

$$\therefore \text{L.H.L} \neq \text{R.H.L}$$

$\Rightarrow$ limit does not exist

$\Rightarrow f$ is not cts

Graphically,



Clearly,

Graph of given fth has a gap (jump) ~~at~~

$\Rightarrow$ So, function is not continuous.

$$f(x) = \begin{cases} a+xb & , & x < 1 \\ a-xb & , & x > 1 \\ 1 & , & x = 1 \end{cases}$$

43

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} a+xb = a+b.$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} a-xb = a-b.$$

Since function is cts at $x=1$

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

$$a+b = a-b = 1$$

Solve simultaneously.

Adding

$$\begin{array}{r} a+b=1 \\ a-b=1 \\ \hline 2a=2 \\ \boxed{a=1} \end{array}$$

Sub

$$\begin{array}{r} a+b=1 \\ + a-b=1 \\ \hline - + - \\ 2b=0 \\ \boxed{b=0} \end{array}$$

$$a=1; b=0.$$

Q32- $f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

Ex No 1.9
Q(1-43)
For Practice

Sol:

$$\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) = \alpha \in [-1, 1].$$

$$\therefore f(0) = 0.$$

$f(x)$ is cts, when $\lim_{x \rightarrow 0} f(x) = f(0)$

$$\text{i.e. } \lim_{x \rightarrow 0} \sin \frac{1}{x} = 0.$$

$\Rightarrow$ Given ftn is cts when $x=0$, otherwise for $x \neq 0$ f is discontinuous.

Q- Q3 $T(x) = \lfloor 1-x \rfloor + \lfloor x-1 \rfloor.$

check the continuity of $T(x)$ at $x=1$.

$$T(1) = \lfloor 1-1 \rfloor + \lfloor 1-1 \rfloor = \lfloor 0 \rfloor + \lfloor 0 \rfloor = 0.$$

$$\lim_{x \rightarrow 1^-} T(x) = \lfloor 1-0.999 \rfloor + \lfloor 0.999-1 \rfloor$$

$$= 0$$

$$= 0 - 1 = -1$$

$$\lim_{x \rightarrow 1^+} T(x) = \lfloor 1-1.0001 \rfloor + \lfloor 1.0001-1 \rfloor$$

$$= \lfloor -0.0001 \rfloor + \lfloor 0.0001 \rfloor$$

$$= -1 + 0$$

$$= -1$$

$$\therefore \lim_{x \rightarrow 1} T(x) \neq T(1)$$

$\Rightarrow$ Ftn is not cts at $x=1$.